浙江大学
ZHEJIANG UNIVERSITY

Recitation 4

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勤学 / 修德 / 明辨 / 笃实



Problem 1

11.2. Consider the interconnection of continuous-time LTI systems shown in Figure P11.2. Express the overall system function for this interconnection in terms of $H_1(s)$, $H_2(s)$, $G_1(s)$, and $G_2(s)$.

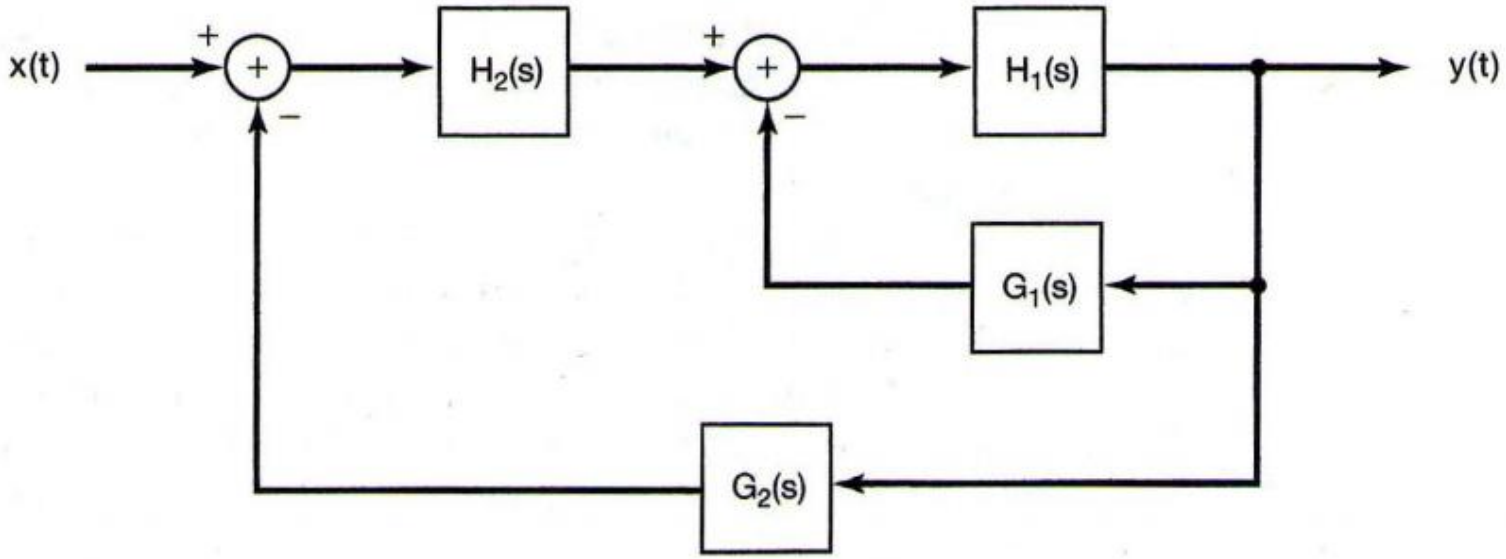


Figure P11.2

Problem 1

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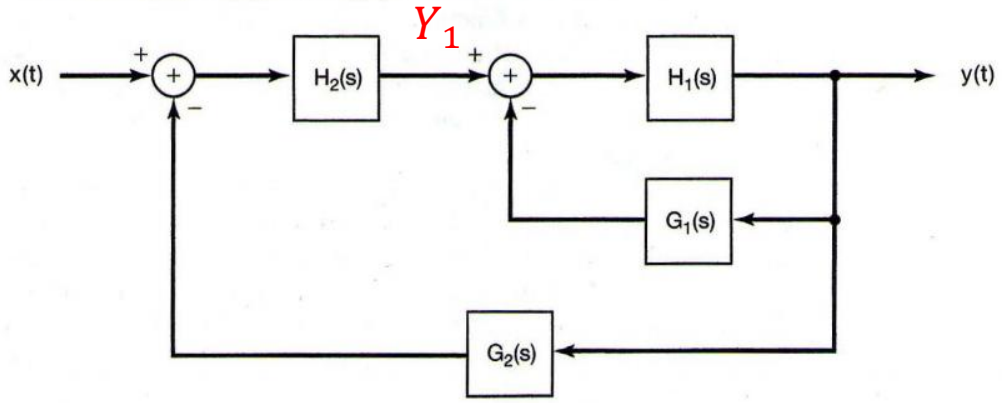


Figure P11.2

$$\begin{cases} Y_1 = H_2(X - G_2Y) \\ Y = H_1(Y_1 - YG_1) \end{cases}$$

$$Y = H_1(H_2(X - G_2Y) - YG_1)$$

$$= H_1(H_2X - H_2G_2Y - G_1Y)$$

$$= H_1H_2X - H_1H_2G_2Y - H_1G_1Y$$

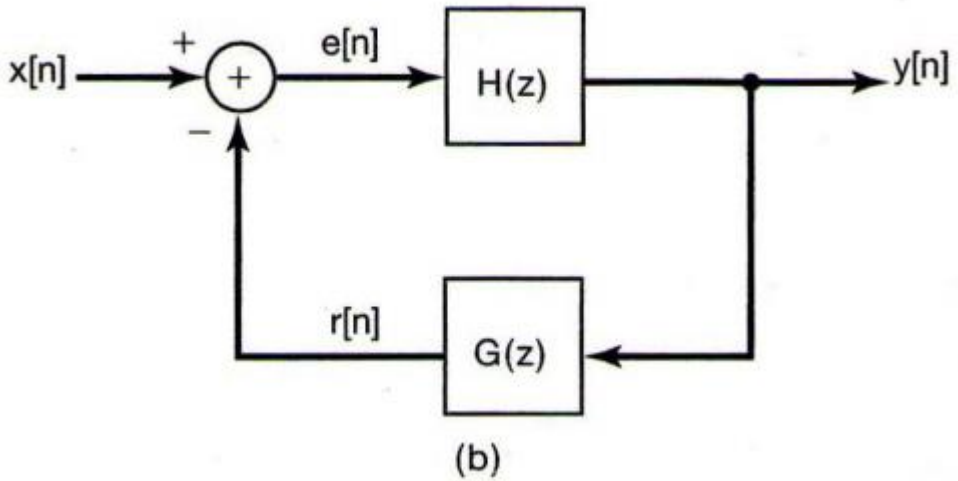
$$\frac{Y}{X} = \frac{H_1H_2}{1 + H_1H_2G_2 + H_1G_1}$$

Problem 2

11.5. Consider the discrete-time feedback system depicted in Figure 11.3(b) with

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad \text{and} \quad G(z) = 1 - bz^{-1}.$$

For what real values of b is the feedback system stable?



(b)

$$Y = H(X - GY)$$

$$\frac{Y}{X} = \frac{H}{1+HG} = \frac{\frac{1}{1-\frac{1}{2}z^{-1}}}{1+\frac{1-bz^{-1}}{1-\frac{1}{2}z^{-1}}} = \frac{1}{2-\left(\frac{1}{2}+b\right)z^{-1}}$$

$$z_0 = \frac{1+2b}{4}$$

$$|z_0| = \left|\frac{1+2b}{4}\right| < 1$$

$$-\frac{5}{2} < b < \frac{3}{2}$$

因果：收敛域包含无穷远
稳定：收敛域包含单位圆
因果稳定：所有极点都在单位圆内

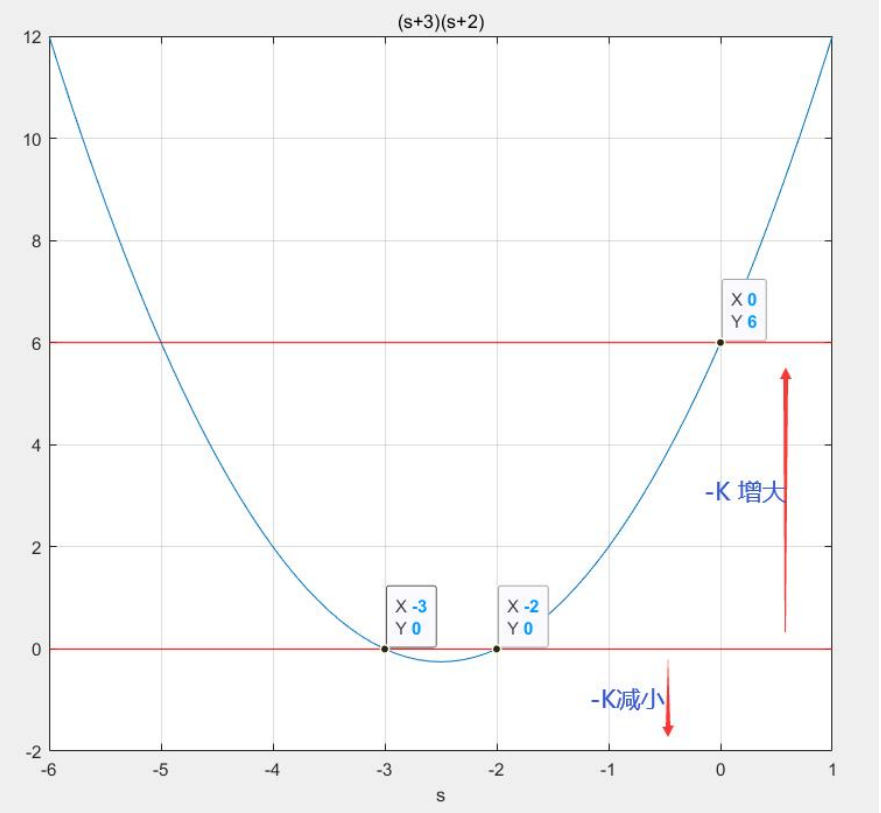
Problem 3

11.7. Suppose the closed-loop poles of a feedback system satisfy

$$\frac{1}{(s+2)(s+3)} = -\frac{1}{K}.$$

Use the root-locus method to determine the values of K for which the feedback system is guaranteed to be stable.

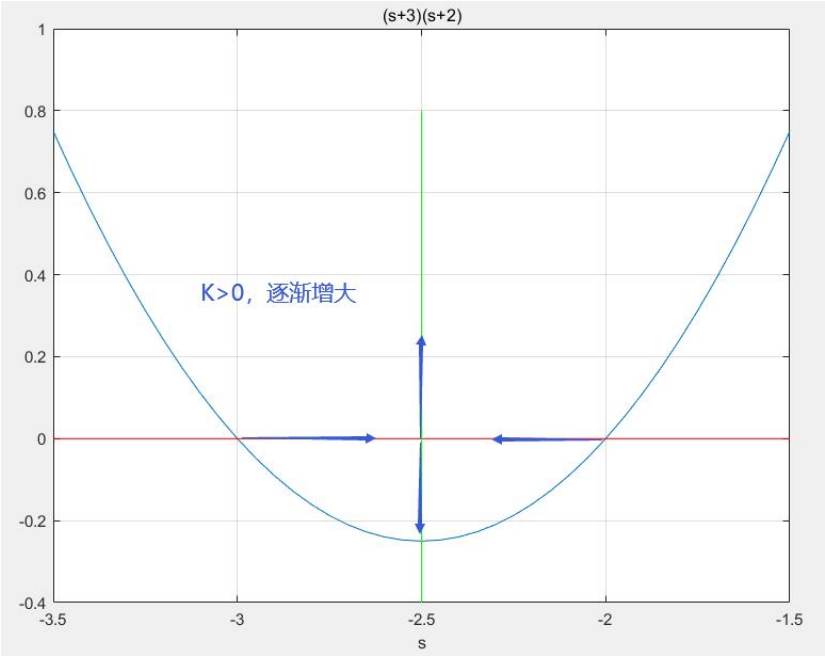
$$(s+2)(s+3) = -k$$



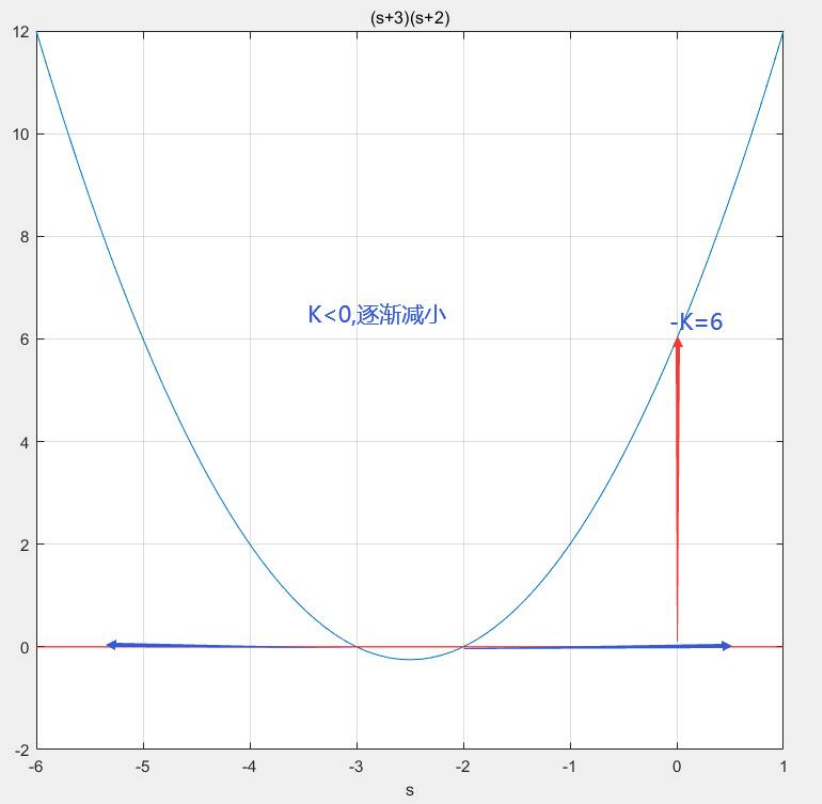
Problem 3

$$(s+2)(s+3) = -k$$

$k > 0$: stable



$k < 0$:



when K<=-6， not stable

Problem 3

$$s^2 + 5s + 6 + k = 0$$

$$s_{0,1} = \frac{-5 \pm \sqrt{25 - 4(6+k)}}{2}$$

Non-stable

$$\frac{-5 + \sqrt{25 - 4(6+k)}}{2} \geq 0$$

$$6 + k \leq 0$$

$$k \leq -6$$

因果：收敛域为复平面的右半边
稳定：收敛域包含虚轴
因果稳定：最右边极点在虚轴的左边

Problem 4

11.57. In this problem, we consider several examples of the design of tracking systems. Consider the system depicted in Figure P11.57. Here, $H_p(s)$ is the system whose output is to be controlled, and $H_c(s)$ is the compensator to be designed. Our objective in choosing $H_c(s)$ is that we would like the output $y(t)$ to follow the input $x(t)$. In particular, in addition to stabilizing the system, we would also like to design the system so that the error $e(t)$ decays to zero for certain specified inputs.

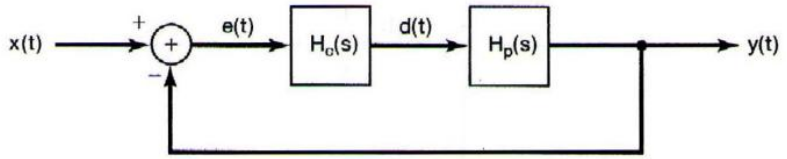


Figure P11.57

(a) Suppose that

$$H_p(s) = \frac{\alpha}{s+\alpha}, \quad \alpha \neq 0. \tag{P11.57–1}$$

Show that if $H_c(s) = K$ (which is known as *proportional* or *P* control), we can choose K so as to stabilize the system and so that $e(t) \to 0$ if $x(t) = \delta(t)$. Show that we *cannot* get $e(t) \to 0$ if $x(t) = u(t)$.

(b) Again let $H_p(s)$ be as in eq. (P11.57–1), and suppose that we use *proportional-plus-integral* (PI) control—that is,

$$H_c(s) = K_1 + \frac{K_2}{s}.$$

Show that we can choose K_1 and K_2 so as to stabilize the system, and we can also get $e(t) \to 0$ if $x(t) = u(t)$. Thus, the system can track a step. In fact, this illustrates a basic and important principle in feedback system design: To track a step [$X(s) = 1/s$], we need an integrator ($1/s$) in the feedback system. An extension of this principle is considered in the next problem.

(c) Suppose that

$$H_p(s) = \frac{1}{(s-1)^2}.$$

Show that we cannot stabilize this system with a PI controller, but that we can stabilize it and have it track a step if we use *proportional-plus-integral-plus-differential* (PID) control, i.e.,

$$H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

Problem 4 a

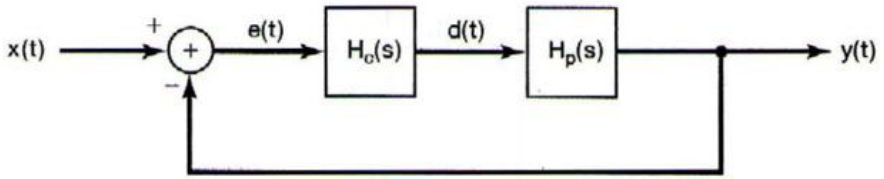


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$$H = \frac{Y}{X} = \frac{H_c H_p}{1 + H_c H_p} = \frac{K\alpha}{s + (1+K)\alpha}$$

$$s_0 = -(1+K)\alpha$$

极点在左半平面

$$-(1+K)\alpha < 0$$

Problem 4 a

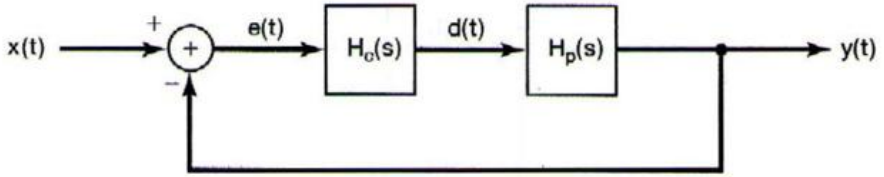


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$$f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

$$H = \frac{Y}{X} = \frac{H_c H_p}{1 + H_c H_p} = \frac{K\alpha}{s + (1+K)\alpha}$$

$$E = X - Y$$

$$\frac{E}{X} = 1 - \frac{Y}{X} = 1 - \frac{K\alpha}{s + (1+K)\alpha}$$

$x(t) = \delta(t), \ X(s) = 1$

$$E(s) = 1 - \frac{K\alpha}{s+(1+K)\alpha}$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\left(1 - \frac{K\alpha}{s + (1+K)\alpha}\right)$$

$$= \lim_{s\to 0}\left(s - \frac{K\alpha s}{s+(1+K)\alpha}\right) = 0$$

$x(t) = u(t), \ X(s) = \frac{1}{s}$

$$E(s) = \frac{1}{s}\left(1 - \frac{K\alpha}{s + (1+K)\alpha}\right)$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0}\left(1 - \frac{K\alpha}{s + (1+K)\alpha}\right)$$

$$= \frac{1}{(1+K)\alpha} \neq 0$$

Problem 4 b

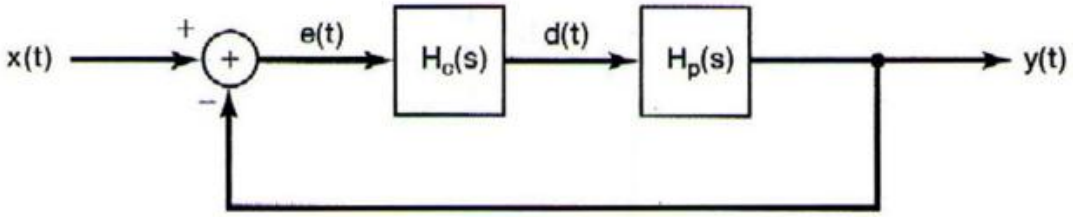


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$$H = \frac{\frac{\alpha}{s+\alpha}\left(K_1 + \frac{K_2}{s}\right)}{1 + \frac{\alpha}{s+\alpha}\left(K_1 + \frac{K_2}{s}\right)} = \frac{\alpha K_1 s + \alpha K_2}{s^2 + \alpha(1+K_1)s + \alpha K_2}$$

$$p_{0,1} < 0$$

$$p_0 + p_1 = \alpha(1+K_1) < 0$$

$$p_0 p_1 = \alpha K_2 > 0$$

Problem 4 b

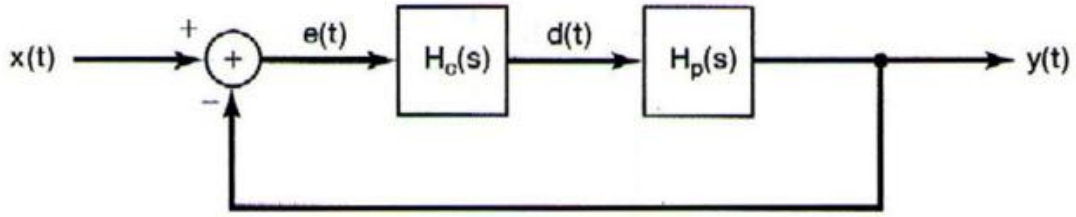


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$$E = 1 - \frac{Y}{X} = 1 - \frac{\alpha K_1 s+\alpha K_2}{s^2+\alpha(1+K_1)s+\alpha K_2}$$

$$x(t) = u(t),\ \ X(s) = \frac{1}{s}$$

$$E(s) = \frac{1}{s}\left(1 - \frac{\alpha K_1 s+\alpha K_2}{s^2+\alpha(1+K_1)s+\alpha K_2}\right)$$

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Problem 4 c

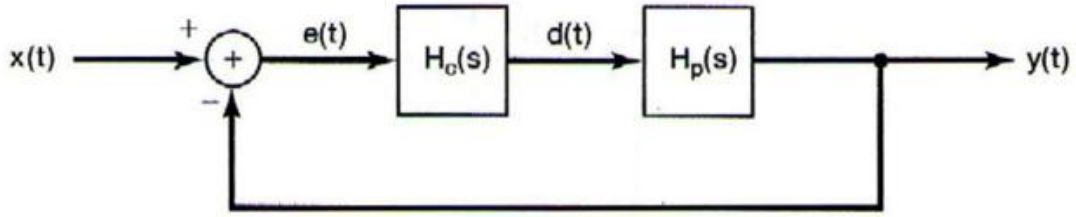


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(c) Suppose that

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$$H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

$$PI: H(s) = \frac{Y(s)}{X(s)} = \frac{(K_1 s + K_2)}{s^3 - 2s^2 + (K_1 + 1)s + K_2}$$

$$(s-p_0)(s-p_1)(s-p_2)$$

$$= s^3 - (p_0 + p_1 + p_2)s^2 + (p_0p_1 + p_0p_2 + p_1p_2)s - p_0p_1p_2$$

$$p_0 + p_1 + p_2 = 2$$

存在右半平面的极点，不可能稳定

Problem 4 c

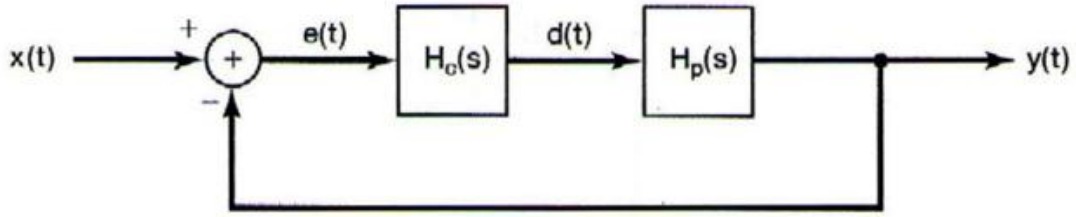


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$$H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

劳斯判据

$D(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0$

s^3	a_3	a_1
s^2	a_2	a_0
s^1	b_1	0
s^0	c_1	0

$$b_1 = \frac{a_2 a_1 - a_3 a_0}{a_2}, \quad c_1 = a_0$$

$$PID: H(s) = \frac{Y(s)}{X(s)} = \frac{(K_3 s^2 + K_1 s + K_2)}{s^3 + (K_3 - 2)s^2 + (K_1 + 1)s + K_2}$$

$$(s-p_0)(s-p_1)(s-p_2)$$

$$= s^3 - (p_0 + p_1 + p_2)s^2 + (p_0 p_1 + p_0 p_2 + p_1 p_2)s - p_0 p_1 p_2$$

$$p_0 < 0, p_1 < 0, p_2 < 0 \longrightarrow K_3 - 2 > 0, K_1 + 1 > 0, K_2 > 0$$

$a_3 > 0$

$a_2 > 0$

$b_1 > 0$

$c_1 > 0$

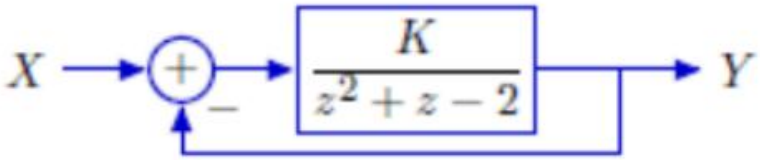
$K_3 - 2 > 0$

$(K_3 - 2)(K_1 + 1) > K_2$

$K_2 > 0$

Problem 5

Consider the following feedback system in which the box represents a causal LTI DT system that is represented by its system function.



a. Determine the range of K for which this feedback system is stable.
b. Determine the range of K for which this feedback system has real-valued poles.

$$H = \frac{\frac{K}{z^2 - z - 2}}{1 + \frac{K}{z^2 - z - 2}} = \frac{K}{z^2 - z - 2 + K}$$

$$z_{0,1} = \frac{1 \pm \sqrt{1 - 4(K-2)}}{2}$$

$1 - 4(K-2) \geq 0$

$K \leq \frac{9}{4}$

$\frac{1+\sqrt{1-4(K-2)}}{2} < 1$

$2 < K$

$2 < K \leq \frac{9}{4}$

$1 - 4(K-2) < 0$

$\frac{9}{4} < K$

$z_{0,1} = \frac{1 \pm j\sqrt{4(K-2) - 1}}{2}$

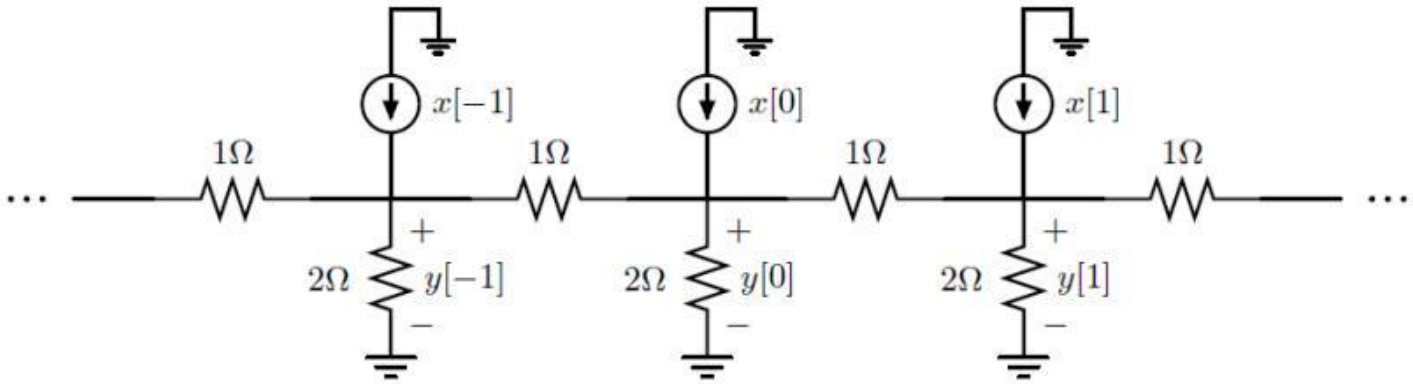
$\sqrt{1 + 4(K-2) - 1} < 2$

$K < 3$

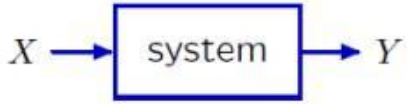
$\frac{9}{4} < K < 3$

Problem 6

An infinite network of resistors is excited by an infinite network of current sources as shown below.



We can consider the transformation from x to y as a DT system.



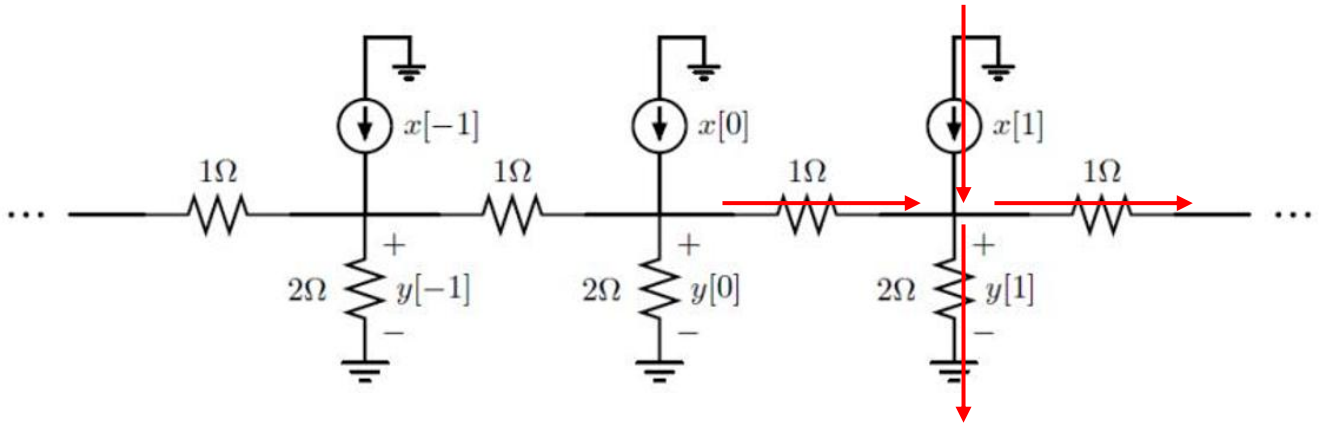
a. Show that this system is linear and "time"-invariant.

b. Determine the unit-sample response $h[n]$.

c. Determine the system function $H(z)$ and region of convergence.

d. Determine the system's pole(s) and zero(s).

Problem 6 a



$$\frac{y[n-1]-y[n]}{1}+x[n]=\frac{y[n]}{2}+\frac{y[n]-y[n+1]}{1}$$

$$x[n]=\frac{5}{2}y[n]-y[n-1]-y[n+1]$$

线性

齐次性：

$x[n]\to y[n]$

$$ax[n]=a\left(\frac{5}{2}y[n]-y[n-1]-y[n+1]\right)$$

$ax[n]\to ay[n]$

可加性：

$x_1[n]\to y_1[n],\ x_2[n]\to y_2[n]$

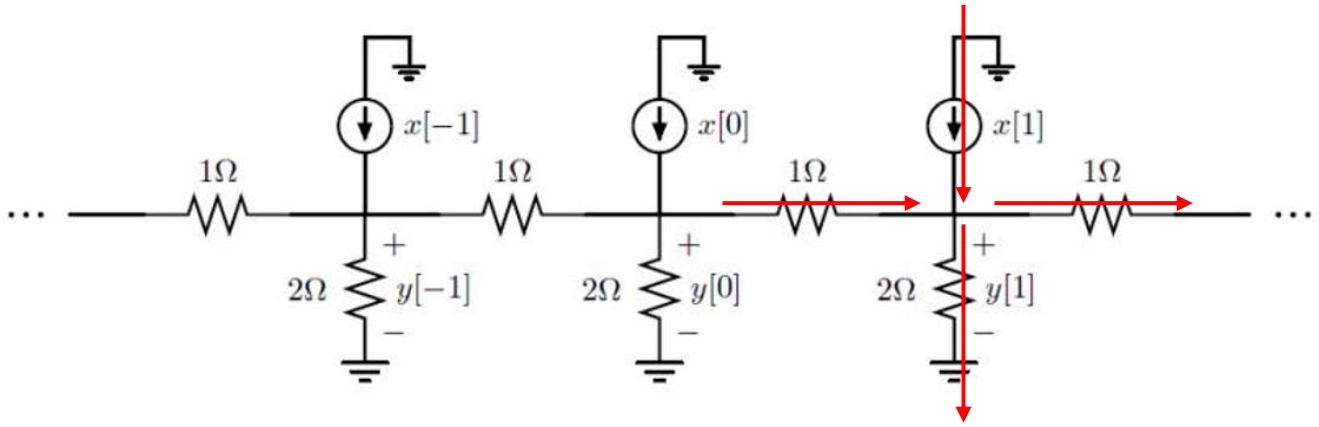
$$x_1[n]=\frac{5}{2}y_1[n]-y_1[n-1]-y_1[n+1]$$

$$x_2[n]=\frac{5}{2}y_2[n]-y_2[n-1]-y_2[n+1]$$

$$x_1[n]+x_2[n]=\frac{5}{2}(y_1[n]+y_2[n])-(y_1[n-1]+y_2[n-1])-(y_1[n+1]+y_2[n+1])$$

$x_1[n]+x_2[n]\to y_1[n]+y_2[n]$

Problem 6 a



时不变：

$x[n] \to y[n]$

$$x[n] = \frac{5}{2}y[n] - y[n-1] - y[n+1]$$

$$x[n-k] = \frac{5}{2}y[n-k] - y[n-k-1] - y[n-k+1]$$

$x[n-k] \to y[n-k]$

$$\frac{y[n-1]-y[n]}{1} + x[n] = \frac{y[n]}{2} + \frac{y[n]-y[n+1]}{1}$$

$$x[n] = \frac{5}{2}y[n] - y[n-1] - y[n+1]$$

Problem 6 b

b. Determine the unit-sample response $h[n]$.

c. Determine the system function $H(z)$ and region of convergence.

d. Determine the system's pole(s) and zero(s).

$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$
$-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert < \lvert\alpha\rvert$

$$x[n] = \frac{5}{2}y[n] - y[n-1] - y[n+1]$$

$$H = \frac{Y}{X} = \frac{1}{\frac{5}{2} - z - z^{-1}} = \frac{-2z}{2z^2 - 5z + 2} = \frac{-2z}{(z-2)(2z-1)}, \quad \frac{1}{2} < |z| < 2$$

$$= -\frac{2}{3}\frac{1}{1-2z^{-1}} + \frac{2}{3}\frac{1}{1-\frac{1}{2}z^{-1}}$$

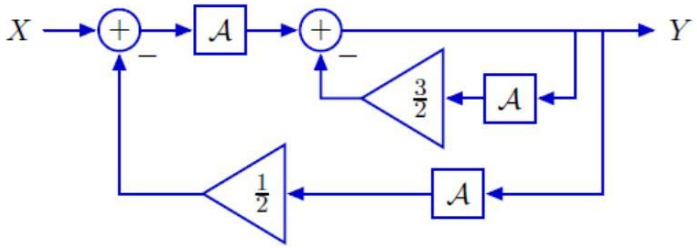
$$h[n] = \frac{2}{3}2^n u[-n-1] + \frac{2}{3}\left(\frac{1}{2}\right)^n u[n] = \frac{2}{3}\left(\frac{1}{2}\right)^{|n|}$$

$$z_0 = 0$$

$$p_0 = 2, p_1 = \frac{1}{2}$$

HW 2 P-1-b

Consider the system defined by the following block diagram:



a. Determine the system functional $H = \frac{Y}{X}$.

b. Determine the poles of the system.

c. Determine the impulse response of the system.

$$H = \frac{2A}{2 + 3A + A^2}$$

极点是指拉普拉斯变换的极点

$$H = \frac{\frac{2}{s}}{\left(1 + \frac{1}{s}\right)\left(2 + \frac{1}{s}\right)}$$

$$= \frac{2s}{(s+1)(2s+1)}$$

$$= \frac{2}{s+1} - \frac{2}{2s+1}$$

$$p_0 = -1, p_1 = -\frac{1}{2}$$

HW 2 P-2-b

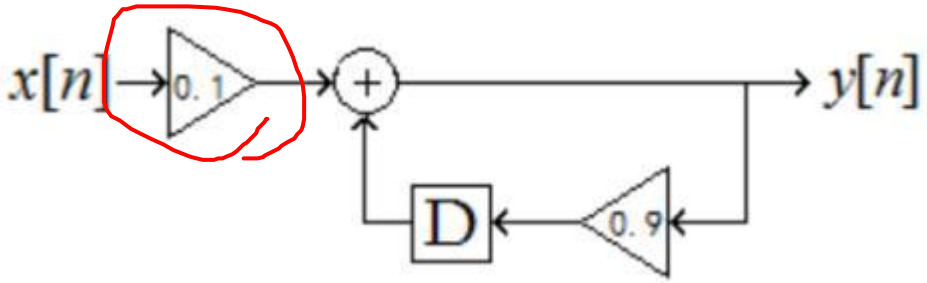
a. Determine the difference equation and block diagram representations for a system whose output is 10, 1, 1, 1, 1, … when the input is 1, 1, 1, 1, 1, … .

b. Determine the difference equation and block diagram representations for a system whose output is 1, 1, 1, 1, 1, . . . when the input is 10, 1, 1, 1, 1, ...

c. Compare the difference equations in parts a and b. Compare the block diagrams in parts a and b.

x[n] = 10y[n] – 9y[n-1]

y[n] = 0.1x[n] + 0.9y[n-1]



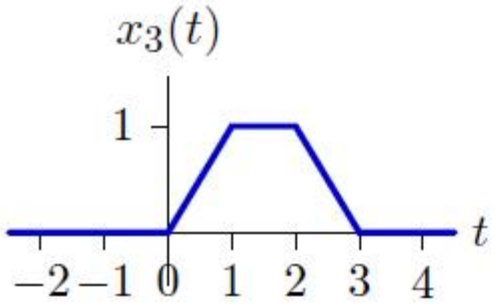
HW 2 P-3-c

Determine the Laplace transforms (including the regions of convergence) of each of the following signals:

a. $x_1(t) = e^{-2(t-3)}u(t-3)$

b. $x_2(t) = |t|e^{-|t|}$

c.



$$X_3(s) = \int_0^1 t\,e^{-st}dt + \int_1^2 e^{-st}\,dt + \int_2^3 (3-t)\,e^{-st}dt$$

$$= \left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right)\Big|_0^1 - \left(\frac{e^{-st}}{s}\right)\Big|_1^2 - \left(\frac{3e^{-st}}{s}\Big|_2^3 - \left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right)\Big|_2^3\right)$$

$$= \frac{e^{-3s} - e^{-2s} - e^{-s} + 1}{s^2}$$

在s=0的极限存在，没有极点，ROC为全复平面

HW 2 P-4-b

Determine and sketch all possible signals with Laplace transforms of the following forms.
For each signal, indicate the associated region of convergence.

a. $X_1(s) = \dfrac{s+2}{(s+1)^2}$

b. $X_2(s) = \left(\dfrac{1-e^{-s}}{s}\right)^2$

$$X_2(s) = \frac{1-2e^{-s}+e^{-2s}}{s^2}$$

由于s->0的时候，$X_2(s)$是存在的，所以并没有极点，可以不讨论ROC。

$$Re\{s\} > 0, \quad tu(t) \Leftrightarrow \frac{1}{s^2}, \qquad (t-1)u(t-1) \Leftrightarrow \frac{e^{-s}}{s^2}, \quad (t-2)u(t-2) \Leftrightarrow \frac{e^{-2s}}{s^2}$$

$$X_2(s) = tu(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$$

$$Re\{s\} < 0, \quad -tu(-t) \Leftrightarrow \frac{1}{s^2}, \quad -(t-1)u(1-t) \Leftrightarrow \frac{e^{-s}}{s^2}, \quad -(t-2)u(2-t) \Leftrightarrow \frac{e^{-2s}}{s^2}$$

$$X_2(s) = -tu(-t) + 2(t-1)u(1-t) - (t-2)u(2-t)$$

实际上是同一个函数

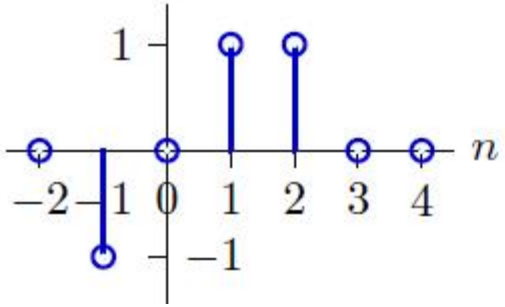
HW 2 P-7-c

Determine the Z transform (including the region of convergence) for each of the following signals:

a. $x_1[n] = \left(\frac{1}{2}\right)^n u[n-3]$

b. $x_2[n] = (1+n)\left(\frac{1}{3}\right)^n u[n]$

c. $x_3[n]$



$$X_3(z) = \sum_{n=-\infty}^{\infty} x_3[n] z^{-n}$$

$$= -z + z^{-1} + z^{-2}$$

ROC: ALL z except 0 and ∞

HW 2 P-9-a

a. Under what conditions does the Z transform of $x_e[n]$ exist?

$$x_e[n] = \frac{x[n]+x[-n]}{2}$$

Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least the intersection of R_1 and R_2

$$X_e(z) = \frac{X(z)+X(z^{-1})}{2}$$

$X(z)$ $\quad ROC: r_0 < |z| < r_1$

$X(z^{-1})$ $\quad ROC: \frac{1}{r_1} < |z| < \frac{1}{r_0}$

$X_e(z)$ $\quad ROC: \max\left(r_0, \frac{1}{r_1}\right) < |z| < \min\left(r_1, \frac{1}{r_0}\right)$

$\Rightarrow r_1 > \frac{1}{r_1},\ r_0 < \frac{1}{r_0}$

$\Rightarrow r_0 < 1 < r_1$

HW 2 P-10-a

9.28. Consider an LTI system for which the system function $H(s)$ has the pole-zero pattern shown in Figure P9.28.

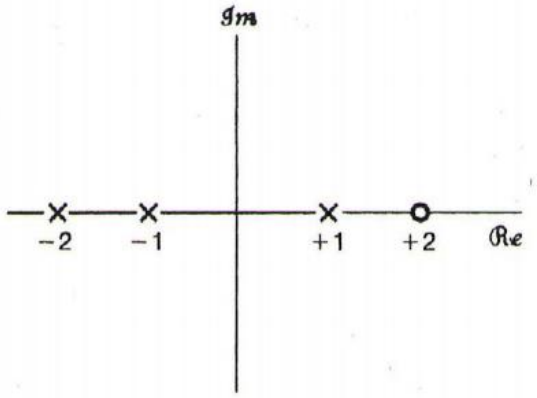


Figure P9.28

(a) Indicate all possible ROCs that can be associated with this pole-zero pattern.
(b) For each ROC identified in part (a), specify whether the associated system is stable and/or causal.

	Casual	Stable
1) $\mathrm{Re}\{s\} > 1$	√	×
2) $-1 < \mathrm{Re}\{s\} < 1$	×	√
3) $-2 < \mathrm{Re}\{s\} < -1$	×	×
4) $\mathrm{Re}\{s\} < -2$	×	×

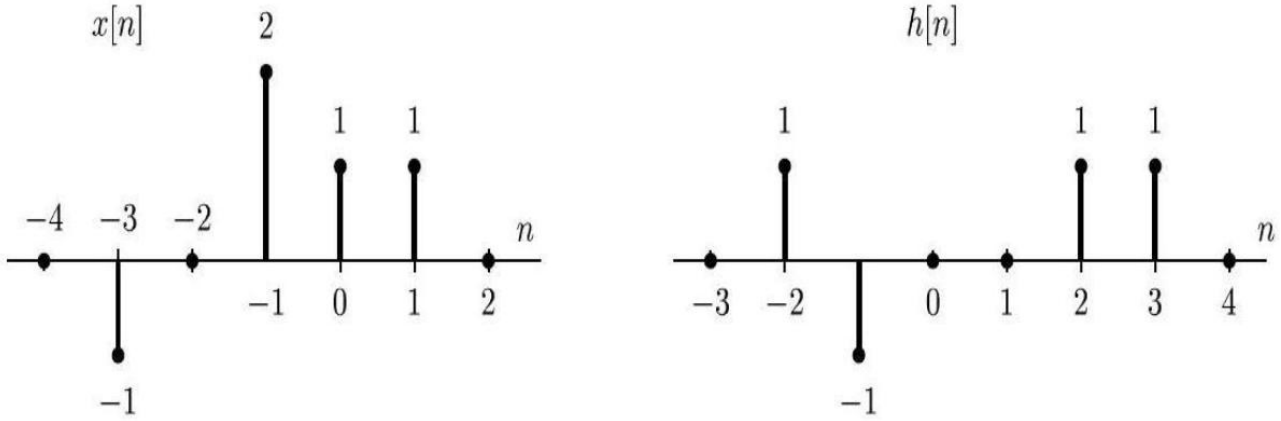
ROC不能有极点

LTI

Casual ⇔ ROC is **right-side**

Stable ⇔ ROC contains **imaginary axis**

HW 3 P-1-a



$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$

$\{1\ -1\ 0\ 0\ 1\ 1\}_{-2}$ $h[n]$

$\{-1\ 0\ 2\ 1\ 1\}_{-3}$ $x[n]$

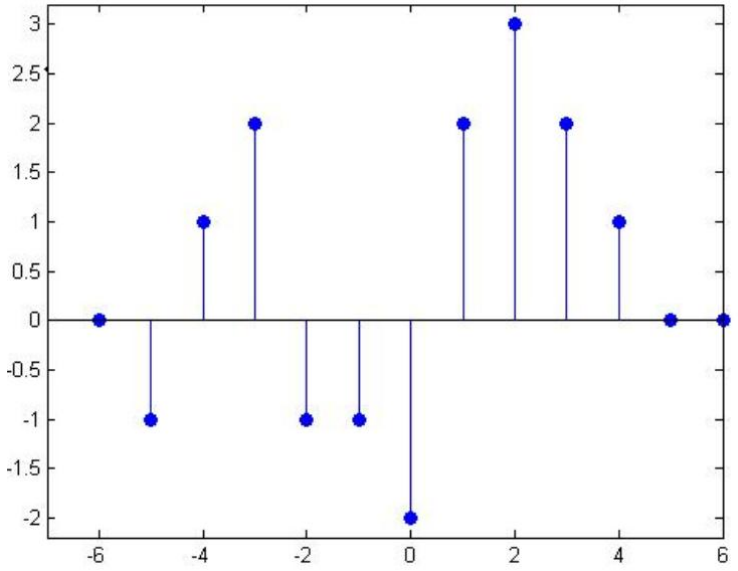
$-1\ 1\ 0\ 0\ -1\ -1$ $x[-3]\,h[n+3]$

$2\ -2\ 0\ 0\ 2\ 2$ $x[-1]\,h[n+1]$

$1\ -1\ 0\ 0\ 1\ 1$ $x[0]\,h[n]$

$1\ -1\ 0\ 0\ 1\ 1$ $x[1]\,h[n-1]$

$\{-1\ 1\ 2\ -1\ -1\ -2\ 2\ 3\ 2\ 1\}_{-5}$



HW 3 P-1-b

(b). $x[n] = u[n+4] - u[n-1], h[n] = 2^n u[2-n].$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{2} 2^k u[2-k](u[n-k+4] - u[n-k-1]) = \sum_{k=n}^{n+4} 2^k u[2-k] = \begin{cases} 31 \times 2^n, & n \le -2 \\ 7.5, & n = -1 \\ 7, & n = 0 \\ 6, & n = 1 \\ 4, & n = 2 \\ 0, & else \end{cases}$$

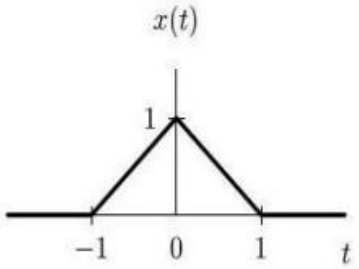
[n, n+4]∩[-∞, 2]

1. n <= -2，积分区间[n, n+4]

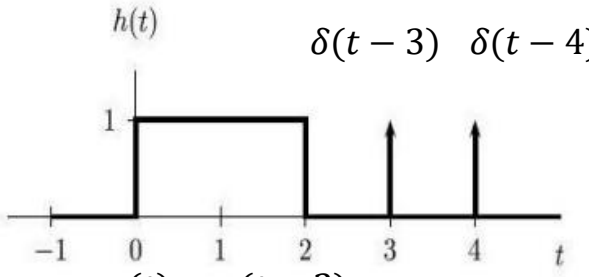
2. -2 < n <=2，积分区间[n, 2]

3. 2 < n，没有交集，结果为0

HW 3 P-2-b



*



$\delta(t-3)$ $\delta(t-4)$

$u(t)-u(t-2)$

$$x(t)=\begin{cases}t+1, -1\le t<0\\ 1-t, 0\le t<1\end{cases}$$

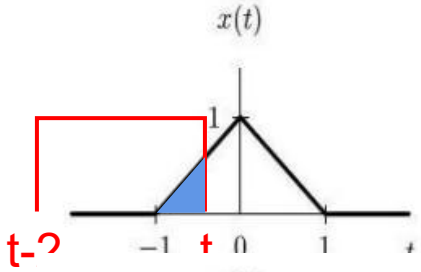
$$y_1(t)=\begin{cases}t-2, 2\le t<3\\ 4-t, 3\le t<4\end{cases}$$

$$y_2(t)=\begin{cases}t-3, 3\le t<4\\ 5-t, 4\le t<5\end{cases}$$

$$y(t)=x(t-3)+x(t-4)+\int_{t-2}^{t}x(\tau)d\tau$$

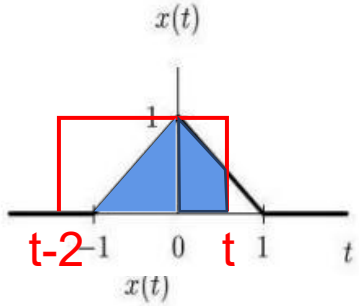
HW 3 P-2-b

1



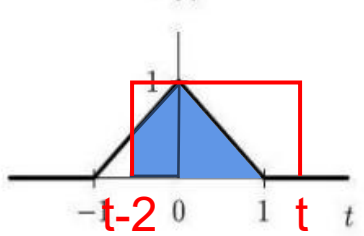
$$y_3(t) = \frac{1}{2}(t+1)^2, -1 \le t < 0$$

2



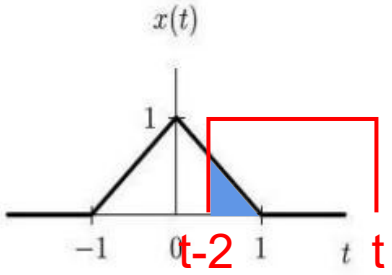
$$y_3(t) = \frac{1}{2} + \frac{(2-t)t}{2} = -\frac{1}{2}t^2 + t + \frac{1}{2}, 1 \le t < 0$$

3



$$y_3(t) = \frac{1}{2} + \frac{(2-t)(1+t-1)}{2} = -\frac{1}{2}t^2 + t + \frac{1}{2}, 1 \le t < 2$$

4



$$y_3(t) = \frac{(3-t)^2}{2}, 2 \le t < 3$$

HW 3 P-2-b

$$y(t) = x(t-3) + x(t-4) + \int_{t-2}^{t} x(\tau)d\tau$$

$$= \begin{cases} \frac{1}{2}(1+t)^2, & -1 \le t < 0 \\ -\frac{1}{2}t^2 + t + \frac{1}{2}, & 0 \le t < 2 \\ \frac{1}{2}t^2 - 2t + \frac{5}{2}, & 2 \le t < 3 \\ 1, & 3 \le t < 4 \\ -t + 5, & 4 \le t < 5 \\ 0, & else \end{cases}$$

HW 3 P-9

在对数坐标下（波特图），幅值是对数形式：

$$20\log|H(j\omega)|$$

当 ω 远小于任何零点和极点时，系统的增益近似于常数 $20\log K$。

当频率增加并接近某个零点或极点时，它们会改变系统的斜率：

- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。

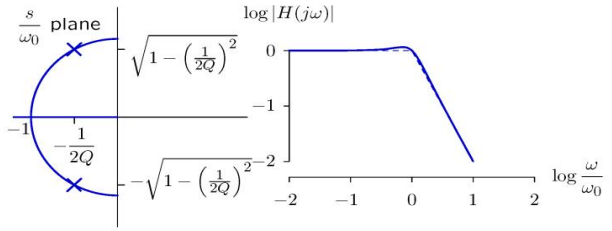
如果有**二阶极点**，则下降率是 40 dB/dec，如果有**二阶零点**，则上升率是 40 dB/dec。

相频图的相位变化不像幅频图那样严格在零极点位置拐折，而是一个**缓慢过渡**：

- **零点（Zero）** 使相位**变化 ±90°**，但相位变化在 $0.1z$ 到 $10z$ 之间分布，而不是严格在 z 处突变。
- **极点（Pole）** 使相位**变化 ±90°**，但相位变化在 $0.1p$ 到 $10p$ 之间分布，而不是严格在 p 处突变。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。
- 相位的变化幅度取决于**零极点的阶数**：
  - **单个零点/极点：变化** $\pm90°$。
  - **二阶零点/极点：变化** $\pm180°$。

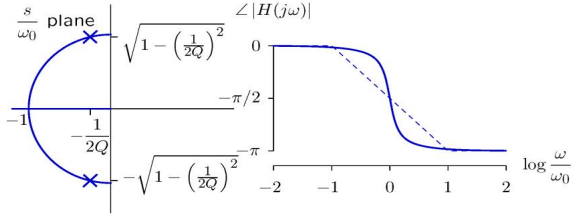
The frequency-response magnitude of a high-Q system is peaked.

$$H(s)=\frac{1}{1+\frac{1}{Q}\frac{s}{\omega_0}+\left(\frac{s}{\omega_0}\right)^2}$$



As Q increases, the phase changes more abruptly with ω.

$$H(s)=\frac{1}{1+\frac{1}{Q}\frac{s}{\omega_0}+\left(\frac{s}{\omega_0}\right)^2}$$



$$H(j\omega)=\frac{1}{1-\left(\frac{\omega}{\omega_0}\right)^2+j\frac{\omega}{Q\omega_0}}.$$

$$\left.\frac{d(\angle H(j\omega))}{d\omega}\right|_{\omega=\omega_0}=-\frac{2Q}{\omega_0},$$

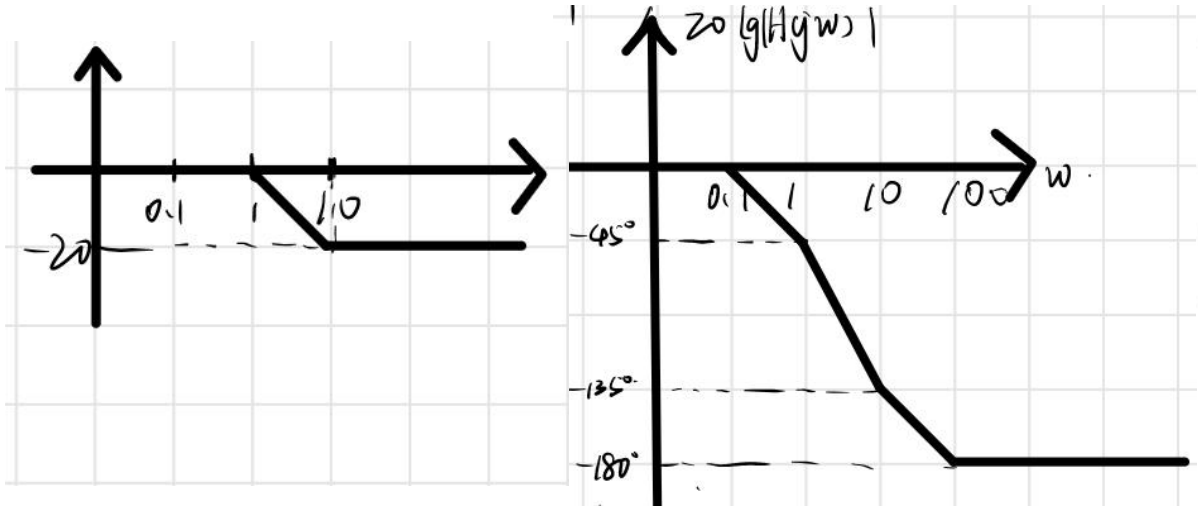
共轭零极点，看作二阶零极点

转折点位于$\omega=\omega_0$处，幅度图转折点出现波峰/波谷，相位图转折点斜率与Q相关

HW 3 P-9-iv-v

(iv) $\dfrac{1-(j\omega/10)}{1+j\omega}$

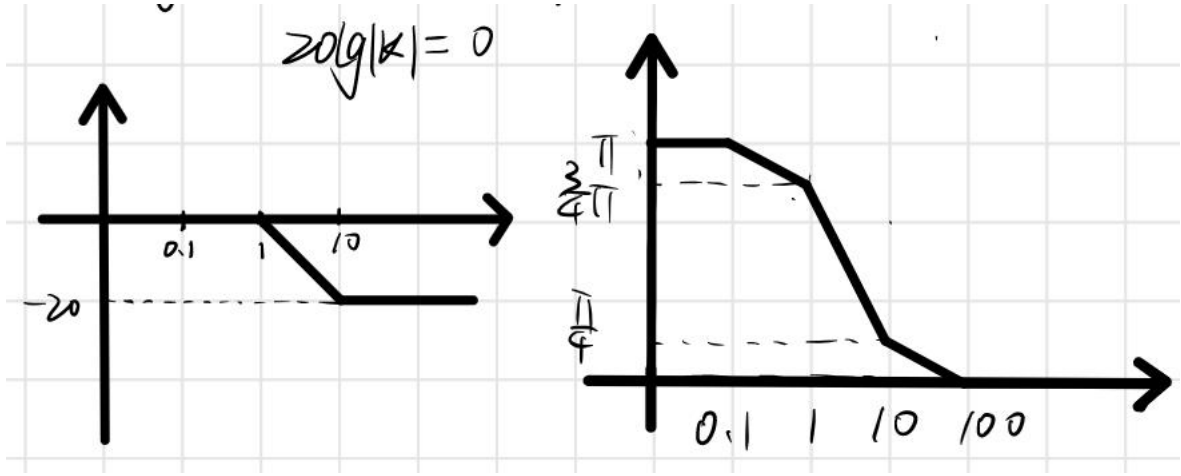
$z_0 = 10$
$p_0 = -1$



- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。

(v) $\dfrac{(j\omega/10)-1}{1+j\omega}$

$z_0 = 10$
$p_0 = -1$



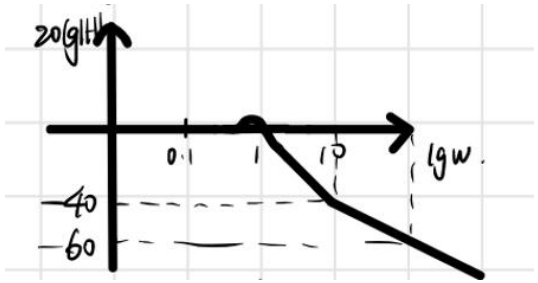
HW 3 P-9-vii-xi

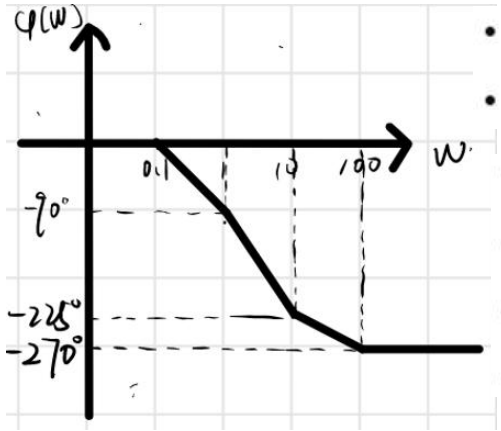
(vii) $\dfrac{1-(j\omega/10)}{(j\omega)^2+(j\omega)+1}$

$z = 10$

$\mathrm{Re}\{p\} = -\dfrac{1}{2}$

$\omega_0 = 1$





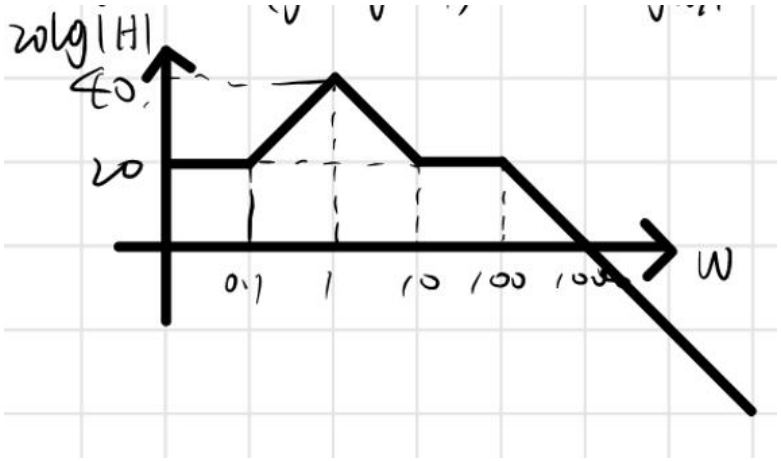
- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。

(xi) $\dfrac{(j\omega+10)(10j\omega+1)}{[(j\omega/100+1)][((j\omega)^2+j\omega+1)]}$

$z_0 = -10,\ z_1 = -0.1$

$p_0 = -100,\ \mathrm{Re}\{p_1\} = -\dfrac{1}{2}$

$\omega_0 = 1$



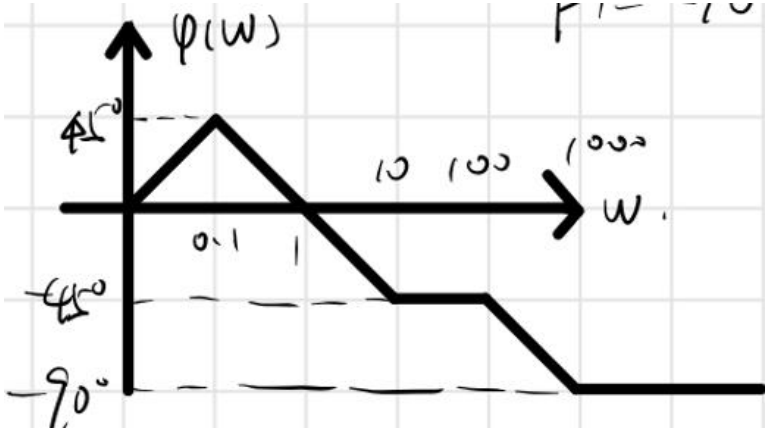


TABLE 9.1 PROPERTIES OF THE LAPLACE TRANSFORM

Section	Property	Signal	Laplace Transform	ROC
		$x(t)$	$X(s)$	R
		$x_1(t)$	$X_1(s)$	R_1
		$x_2(t)$	$X_2(s)$	R_2
9.5.1	Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	At least $R_1 \cap R_2$
9.5.2	Time shifting	$x(t - t_0)$	$e^{-st_0}X(s)$	R
9.5.3	Shifting in the s-Domain	$e^{s_0 t}x(t)$	$X(s - s_0)$	Shifted version of R (i.e., s is in the ROC if $s - s_0$ is in R)
9.5.4	Time scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$	Scaled ROC (i.e., s is in the ROC if s/a is in R)
9.5.5	Conjugation	$x^*(t)$	$X^*(s^*)$	R
9.5.6	Convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	At least $R_1 \cap R_2$
9.5.7	Differentiation in the Time Domain	$\frac{d}{dt}x(t)$	$sX(s)$	At least R
9.5.8	Differentiation in the s-Domain	$-tx(t)$	$\frac{d}{ds}X(s)$	R
9.5.9	Integration in the Time Domain	$\int_{-\infty}^{t} x(\tau)d(\tau)$	$\frac{1}{s}X(s)$	At least $R \cap \{\Re e\{s\} > 0\}$
		Initial- and Final-Value Theorems		
9.5.10	If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses or higher-order singularities at $t = 0$, then $x(0^+) = \lim_{s \to \infty} sX(s)$. If $x(t) = 0$ for $t < 0$ and $x(t)$ has a finite limit as $t \longrightarrow \infty$, then $\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s)$			

TABLE 9.2 LAPLACE TRANSFORMS OF ELEMENTARY FUNCTIONS

Transform pair	Signal	Transform	ROC
1	$\delta(t)$	1	All s
2	$u(t)$	$\frac{1}{s}$	$\Re e\{s\} > 0$
3	$-u(-t)$	$\frac{1}{s}$	$\Re e\{s\} < 0$
4	$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$	$\Re e\{s\} > 0$
5	$-\frac{t^{n-1}}{(n-1)!}u(-t)$	$\frac{1}{s^n}$	$\Re e\{s\} < 0$
6	$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$	$\Re e\{s\} > -\alpha$
7	$-e^{-\alpha t}u(-t)$	$\frac{1}{s+\alpha}$	$\Re e\{s\} < -\alpha$
8	$\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t)$	$\frac{1}{(s+\alpha)^n}$	$\Re e\{s\} > -\alpha$
9	$-\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(-t)$	$\frac{1}{(s+\alpha)^n}$	$\Re e\{s\} < -\alpha$
10	$\delta(t - T)$	e^{-sT}	All s
11	$[\cos \omega_0 t]u(t)$	$\frac{s}{s^2 + \omega_0^2}$	$\Re e\{s\} > 0$
12	$[\sin \omega_0 t]u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$	$\Re e\{s\} > 0$
13	$[e^{-\alpha t}\cos \omega_0 t]u(t)$	$\frac{s+\alpha}{(s+\alpha)^2 + \omega_0^2}$	$\Re e\{s\} > -\alpha$
14	$[e^{-\alpha t}\sin \omega_0 t]u(t)$	$\frac{\omega_0}{(s+\alpha)^2 + \omega_0^2}$	$\Re e\{s\} > -\alpha$
15	$u_n(t) = \frac{d^n\delta(t)}{dt^n}$	s^n	All s
16	$u_{-n}(t) = \underbrace{u(t) * \cdots * u(t)}_{n \text{ times}}$	$\frac{1}{s^n}$	$\Re e\{s\} > 0$

TABLE 10.1 PROPERTIES OF THE z-TRANSFORM

Section	Property	Signal	z-Transform	ROC
		$x[n]$	$X(z)$	R
		$x_1[n]$	$X_1(z)$	R_1
		$x_2[n]$	$X_2(z)$	R_2
10.5.1	Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least the intersection of R_1 and R_2
10.5.2	Time shifting	$x[n - n_0]$	$z^{-n_0}X(z)$	R, except for the possible addition or deletion of the origin
10.5.3	Scaling in the z-domain	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	R
		$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$z_0 R$
		$a^n x[n]$	$X(a^{-1}z)$	Scaled version of R (i.e., $\lvert a\rvert R$ = the set of points $\{\lvert a\rvert z\}$ for z in R)
10.5.4	Time reversal	$x[-n]$	$X(z^{-1})$	Inverted R (i.e., R^{-1} = the set of points z^{-1}, where z is in R)
10.5.5	Time expansion	$x_{(k)}[n] = \begin{cases} x[r], & n = rk \\ 0, & n \neq rk \end{cases}$ for some integer r	$X(z^k)$	$R^{1/k}$ (i.e., the set of points $z^{1/k}$, where z is in R)
10.5.6	Conjugation	$x^*[n]$	$X^*(z^*)$	R
10.5.7	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
10.5.7	First difference	$x[n] - x[n-1]$	$(1 - z^{-1})X(z)$	At least the intersection of R and $\lvert z\rvert > 0$
10.5.7	Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1 - z^{-1}}X(z)$	At least the intersection of R and $\lvert z\rvert > 1$
10.5.8	Differentiation in the z-domain	$nx[n]$	$-z\frac{dX(z)}{dz}$	R
10.5.9		Initial Value Theorem If $x[n] = 0$ for $n < 0$, then $x[0] = \lim_{z\to\infty} X(z)$		

TABLE 10.2 SOME COMMON z-TRANSFORM PAIRS

Signal	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$\lvert z\rvert > 1$
3. $-u[-n-1]$	$\frac{1}{1 - z^{-1}}$	$\lvert z\rvert < 1$
4. $\delta[n - m]$	z^{-m}	All z, except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $\alpha^n u[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$
6. $-\alpha^n u[-n-1]$	$\frac{1}{1 - \alpha z^{-1}}$	$\lvert z\rvert < \lvert\alpha\rvert$
7. $n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$\lvert z\rvert > \lvert\alpha\rvert$
8. $-n\alpha^n u[-n-1]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$\lvert z\rvert < \lvert\alpha\rvert$
9. $[\cos\omega_0 n]u[n]$	$\frac{1 - [\cos\omega_0]z^{-1}}{1 - [2\cos\omega_0]z^{-1} + z^{-2}}$	$\lvert z\rvert > 1$
10. $[\sin\omega_0 n]u[n]$	$\frac{[\sin\omega_0]z^{-1}}{1 - [2\cos\omega_0]z^{-1} + z^{-2}}$	$\lvert z\rvert > 1$
11. $[r^n\cos\omega_0 n]u[n]$	$\frac{1 - [r\cos\omega_0]z^{-1}}{1 - [2r\cos\omega_0]z^{-1} + r^2z^{-2}}$	$\lvert z\rvert > r$
12. $[r^n\sin\omega_0 n]u[n]$	$\frac{[r\sin\omega_0]z^{-1}}{1 - [2r\cos\omega_0]z^{-1} + r^2z^{-2}}$	$\lvert z\rvert > r$